

Japman Singh

+91 79862-12038 | japman006@gmail.com | linkedin.com/in/japman-singh11 | github.com/iJapmanSingh

EXPERIENCE

iHub - AWaDH @ IIT Ropar

June – July 2025

Software Intern — Project: Soil Spectra

IIT Ropar

- Preprocessed a 19,000+ sample hyperspectral soil reflectance dataset (1,000 spectral features, 400–940 nm) using Python, Pandas, and NumPy — performed duplicate removal, outlier treatment, missing value handling, and spectral scaling
- Optimized interpolation step size (0.5 nm, 0.7 nm, etc.) and data cleaning pipelines to maximize R^2 scores of soil nutrient prediction models — achieved R^2 of 0.803 (OC) and R^2 of 0.690 (N) using scikit-learn
- Evaluated multiple regression models to predict soil nutrients (Organic Carbon, Nitrogen, Phosphorus, Potassium) from spectral reflectance data; analyzed RMSE and R^2 across nutrients to identify best-fit configurations
- Collaborated within a cross-functional team (Android + AI/ML) building an end-to-end IoT-based precision agriculture system with BLE-connected optical soil sensors

Annam.ai

June – July 2026

Summer Research Intern

IIT Ropar

- Developed and modified embedded firmware in C for an ultrasonic sensing system on TI MSP430 microcontrollers, using the TI Ultrasonic Sensing Solution (USS) library
- Configured ultrasonic transducers and measured Time-of-Flight (ToF) and Delta-ToF to estimate wind speed and wind direction
- Implemented averaging and filtering of sensor readings to improve measurement stability and reduce noise
- Transmitted processed sensor data via UART for real-time monitoring and analysis
- Debugged firmware and validated sensor behavior on hardware using Code Composer Studio (CCS)

PROJECTS

DevBoard

Full-Stack Developer Productivity Platform

React, Node.js, Express, MongoDB

- Architected a full-stack MERN platform with a modular MVC backend, exposing 15+ RESTful endpoints to unify task management, notes, and third-party coding-platform data into a single dashboard
- Built data-normalization services aggregating LeetCode, Codeforces, and GitHub APIs into a unified schema, reducing manual profile-checking across 3 platforms into a single real-time dashboard view
- Implemented JWT-based authentication with bcrypt hashing and route-level middleware, securing all user sessions and API endpoints against unauthorized access

NoteStack

Backend REST API for Note Management

HTML, CSS, JavaScript, Node.js, Express.js, MongoDB

- Built a RESTful Notes Management API with full CRUD functionality, modular routing, and schema-based validation using Node.js, Express.js, and MongoDB
- Structured backend using a modular architecture with separate routes, controllers, and database models
- Integrated MongoDB with Mongoose for schema-based data persistence
- Tested API endpoints using Postman, implementing request validation and error handling

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, HTML, CSS

Frameworks & Libraries: React, Node.js, Express.js, Tailwind CSS, pandas, NumPy, scikit-learn

Database: MongoDB

Developer Tools: Git, VS Code, IntelliJ IDEA, Postman, MongoDB Compass

Core Competencies: Data Structures & Algorithms, Object-Oriented Programming

Platforms: LeetCode (leetcode.com/u/im_japmansingh/), Codeforces (codeforces.com/profile/japman006)

Achievements: 300+ DSA problems solved (LeetCode, Codeforces, GFG); LeetCode rating: 1600

EDUCATION

Punjabi University

B.Tech in Computer Science

Patiala, Punjab, India

Aug. 2023 – Aug. 2026